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TRACE

Microsoft Research Derives Dense Turn-Level Rewards for Long-Horizon Agents from Temporal-Difference Changes in Frozen Reference Model Log-Probabilities

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Multi-turn tool-use agents have emerged as a dominant paradigm for complex task completion: the agent writes and executes code, calls APIs, browses the web, edits files, and iterates over dozens to hundreds of steps before producing a final answer. Outcome-only reinforcement learning provides a single binary signal at the end of this trajectory. TRACE (Turn-level Reward Assignment via Credit Estimation) replaces this single sparse signal with dense, turn-by-turn rewards derived from how each tool call changes the model’s probability of reaching the correct answer — without any human annotation of intermediate steps.

AT A GLANCE

Problem: Outcome rewards become sparse, high-variance, and misleading for agents operating over 20–100 tool-call trajectories.

Why it matters: Long-horizon agents are increasingly deployed for software engineering, scientific computing, and complex information retrieval; outcome-only RL trains them poorly.

Method: State values estimated as log-ratio of gold-answer log-probabilities (current vs. frozen reference); turn-level rewards are TD differences between consecutive state values.

Result: ToolBench task completion: 41.3% → 58.7%; WebArena: 19.4% → 28.1%; long-horizon code repair: 31.6% → 47.2%.

Evidence: ToolBench, WebArena, and a 20-step code repair benchmark; no intermediate human annotations used.

The Big Picture

Reinforcement learning for single-turn tasks is straightforward: an answer is either right or wrong, and the reward is binary. For multi-turn agents, the binary outcome arrives only at the end of a trajectory that may span dozens of

tool calls and intermediate reasoning steps. Any single turn — a web search, a code edit, a database query — affects the probability of the final correct answer, but outcome-only RL provides no per-turn signal about which turns were helpful.

The credit assignment problem is further complicated by the misleading nature of failed trajectories: a sequence of mostly correct intermediate steps followed by a single error at the last step receives the same negative reward as a trajectory that was wrong from the start, even though the former’s intermediate actions should be reinforced.

Why Existing Approaches Fall Short

Monte Carlo rollout-based credit assigns turn-level rewards by averaging outcomes over multiple rollouts starting from each intermediate state. This is statistically sound but computationally prohibitive for long trajectories: estimating a reliable MC signal for each of 50 turns in a trajectory requires $50 \times K$ rollouts per training example.

Process reward models (PRMs) learn to score intermediate steps, but require human annotation of intermediate step quality, which is expensive and does not scale to the diverse tool-use environments where agents are deployed.

Hindsight relabeling and advantage-based credit methods improve over uniform assignment but remain high-variance for very long trajectories because the value function must extrapolate many steps into the future.

Core Mechanism

TRACE introduces a principled value function for multi-turn trajectories that requires no additional training: the log-ratio of the gold-answer log-probability under the current policy versus a frozen reference model.

At each state s_t (the trajectory prefix through turn t), the state value is:

$$V(s_t) = \log \frac{p_\theta(y^* | s_t)}{p_{\text{ref}}(y^* | s_t)}$$

where y^* is the known gold answer. This quantity has a natural interpretation: it measures how much more likely the current policy is (relative to the frozen reference) to produce the correct answer given the information available at turn t .

The turn-level reward is the Temporal-Difference change:

$$r_t = V(s_t) - V(s_{t-1})$$

Technical Formulation

A trajectory of T turns decomposes into state transitions: $(s_0, a_1, s_1, a_2, \dots, a_T, s_T)$, where each action a_t is a tool call and s_t is the trajectory prefix after the tool result.

The policy π_θ maps from s_{t-1} to a_t ; the tool environment is deterministic.

The TRACE reward satisfies a telescoping property:

$$\sum_{t=1}^T r_t = V(s_T) - V(s_0)$$

where $V(s_T)$ approximates the outcome reward and $V(s_0)$ is a constant baseline. Thus TRACE distributes the outcome signal across turns in a way that is globally consistent: turns that increase V receive positive rewards, turns that decrease it receive negative rewards, and the sum equals the net outcome advantage.

Log-probabilities $p_\theta(y^* \mid s_t)$ are computed by teacher-forcing the gold answer given the trajectory prefix, a standard operation for autoregressive models that requires one forward pass per turn.

Experimental Findings

Benchmark	Outcome RL	TRACE RL
ToolBench	41.3%	58.7%
WebArena	19.4%	28.1%
Code Repair (>20 steps)	31.6%	47.2%

Credit quality measured against human annotations (Pearson r): TRACE achieves $r = 0.61$ vs. $r = 0.22$ for MC rollout-based credit and $r = 0.09$ for uniform assignment.

Ablations and Interpretation

Uniform credit (assigning $V(s_T)/T$ to each turn):

Performance degrades by 8–12% across benchmarks, confirming that the temporal structure of the TD signal is informative — not just the magnitude of the outcome.

Reference model choice: Using a weaker (smaller) reference model degrades credit quality, suggesting the value of a calibrated log-probability baseline. Using the same model checkpoint before RL (as TRACE prescribes) consistently outperforms alternatives.

Trajectory length: The TRACE advantage grows with trajectory length. For tasks with fewer than 5 tool calls, the improvement over outcome-only RL is <2 percentage points. For tasks with 20+ tool calls, the gap is 15+ percentage points, consistent with the hypothesis that credit assignment difficulty scales with trajectory length.

Negative turn identification: TRACE correctly assigns negative rewards to turns that move the agent away from the goal — detectable as a decrease in state value — enabling the policy to specifically learn to avoid those actions.

Reference

Leitian Tao et al. **TRACE: Turn-level Reward Assignment via Credit Estimation for Long-Horizon Agents**. arXiv:2607.13988, July 2026. <https://arxiv.org/abs/2607.13988>